



KHALIL BEN ALI

— Full-Stack & AI Engineer

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SUMMARY

Full-stack engineer (5+ years) shipping TypeScript platforms, .NET desktop apps and production AI. Primary engineer of an AI operations hub integrating 10 SaaS systems for a US financial firm; sole developer of an AI brand experience for JDE's L'OR. Owns systems end to end, from database schema to WebGL, from prompt design to the print queue.

EXPERIENCE

nBold — Full-Stack & AI Engineer · 2024 → present · remote (Paris HQ)

nBold builds Microsoft Teams workspace-templating and governance SaaS (ISO 27001, SOC 2).

- Designed and built the platform's Teams meeting-intelligence feature end to end (~11,000 lines, 67 files, 321 tests): Microsoft Graph transcript capture with a new app-only auth path, AI summarization, and CRM-aware LLM routing to Teams Adaptive Cards, on durable workflows proven crash-safe.
- Primary engineer of Cuddy Brain, a Teams-embedded ops hub mirroring 10 SaaS systems into PostgreSQL (~115 tables); ~3,860 commits, about 69% of backend history (NestJS, BullMQ/Redis, React, Azure OpenAI).
- Shipped AI call intelligence: calls are transcribed, summarized by Azure OpenAI, and turned into follow-up tasks posted to Teams as Adaptive Cards minutes after hang-up, with single-flight locking and crash recovery.
- Built AI monitoring (obituary adjudication with deterministic checks; address-change detection over public records and client communications) and co-built a LangGraph agent that compiles client reviews from 7 systems.
- Hardened the platform: a shared Azure OpenAI client with timeouts, retries and usage tracking, HMAC webhook verification, secret-scanning CI, caching and N+1 performance work.
- Sole developer of L'OR UNLIMITED REMIX (JDE × HLWN 2026), an AI brand app and event kiosk: Gemini with a deterministic fallback, on-device CLIP vision, a 60fps WebGL fluid engine, offline-first PWA and Bluetooth thermal printing. 129/129 commits, ~24.6k lines, 11 weeks to production.
- Built Cuddy Scanner, a .NET 9/WPF TWAIN document scanner (barcode batch separation, PDF assembly, SharePoint filing) in daily production use.

Freelance — Software Developer · 2023 → 2024

- Children's game app (React Native, Expo, Redux) and a cross-platform travel planner (React Native + MobX, Express/MongoDB backend, React admin).

SolvR — Software Developer · 2023

- Refactored and shipped a cross-platform chat app (React Native, Expo, MobX, Firebase): real-time messaging, secure auth, state architecture.

Tynass IT — Software Developer · 2019 → 2022

- Delivered an AR tourism app, a job-seeker platform, fashion e-commerce and corporate sites (React, Material-UI, Express, MongoDB).

SKILLS

TypeScript · React/Next.js · React Native · NestJS/Node · PostgreSQL · Redis/BullMQ · Azure OpenAI · Gemini · LangGraph · RAG (pgvector) · transformers.js · C#/.NET 9 · WPF · C++20/JUCE · Microsoft Graph & Teams · Salesforce · Docker · GitHub Actions · Vercel

SELECTED SIDE PROJECTS

- HyperMesh DSP & HyperGpu (C++20/JUCE): LAN mesh for real-time DSP offload and a GPU-accelerated plugin host; CI builds for 5 platforms.
- Assurances BIAT KB: explainable RAG chatbot (NestJS, pgvector, Ollama) with cited answers and an RBAC audit trail.

EDUCATION

B.A. Multimedia Communication, Higher Institute of Multimedia Arts of Manouba (ISAMM), 2019

LANGUAGES

Arabic (native) · English (professional) · French (A2-B1) · Italian (A2)